

IT114 Module 2 Java Challenge Scenarios

Student submission

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Worksheet link

<https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-module-2-java-challenge-scenarios/ja692>

Grading link

<https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-module-2-java-challenge-scenarios/grading/ja692>

1 submission warning(s) were acknowledged.

AI disclosure: Completed; 1 entry; 1 declared AI use; 0 declared no AI use; policy v1

Java Scenarios

Complete each starter scenario, prove its source and output separately, and explain the decisions used in the solution.

Scenario 1 source - odd values

Edit ``printOdds`` so each passed array prints only odd values on one line, separated by comma and space, with no leading or trailing comma.

URL prompt

Submit the GitHub URL that opens ``M2/Scenario1.java`` on your ``M2-Homework`` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- ``printOdds`` selects only odd values. (up to -0.35 points)
- ``printOdds`` places comma-space separators only between printed values. (up to -0.35 points)

<https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/M2-Homework/M2/Scenario1.java>

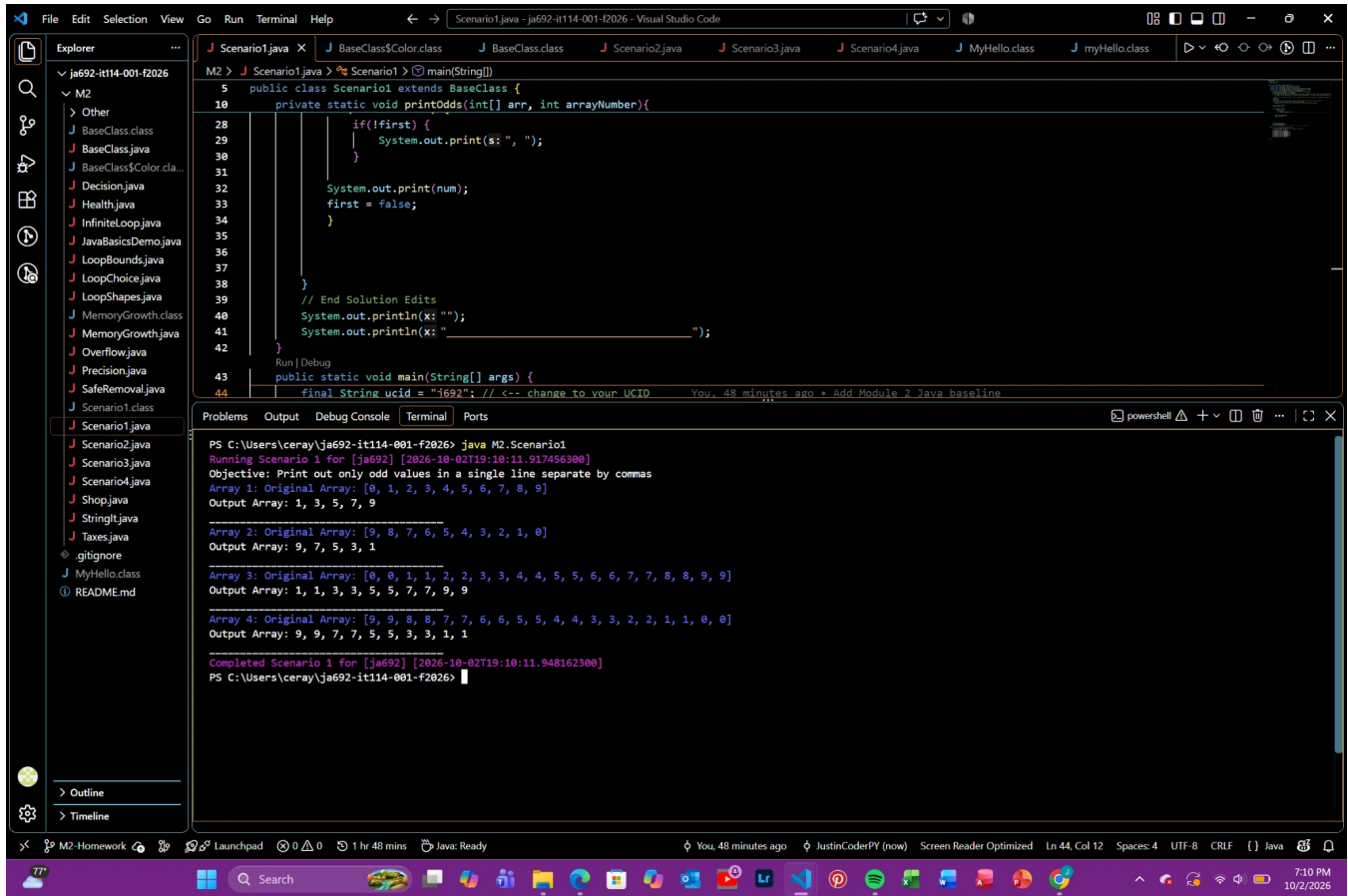
Scenario 1 output

Image evidence prompt

Upload one readable image of the complete ``Scenario1`` program output produced from your ``M2-Homework`` branch.

Requirements and potential deductions

- The output contains only the expected odd array values. (up to -0.35 points)
- Comma-space separators occur only between printed values. (up to -0.2 points)



Scenario 1 explanation

Response prompt

Explain how the code on `M2-Homework` detects odd values and avoids a leading or trailing separator.

Requirements and potential deductions

- The explanation identifies the odd-value test used in the linked code. (up to -0.35 points)
- The explanation identifies the separator decision used in the linked code. (up to -0.2 points)

Submitted response

The code I changed checks odd numbers by using the `num % 2 != 0`, meaning that if that condition is true, those numbers will be printed. From there, the first I created helps keep the commas organized, where every first becomes false, that odd number gets a comma, overall preventing at the beginning or end of the output.

Scenario 2 source - sum and display

Edit `sumValues` to total the passed array in `total` and assign a representation with exactly two decimal places to `modifiedTotal`.

URL prompt

Submit the GitHub URL that opens `M2/Scenario2.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `sumValues` stores the correct raw array sum in `total`. (up to -0.35 points)
- `modifiedTotal` represents the sum with exactly two decimal places. (up to -0.35 points)

<https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/M2-Homework/M2/Scenario2.java>

Scenario 2 output

Image evidence prompt

Upload one readable image of the complete `Scenario2` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output shows the correct computed sums for the starter arrays. (up to -0.35 points)
- The display results have exactly two decimal places. (up to -0.2 points)

```
Scenario2.java - ja692-itt14-001-f2026 - Visual Studio Code
M2 > J Scenario2.java > % Scenario2 > sumValues(double[] int)
5 public class Scenario2 extends BaseClass {
10 private static double[] array5 = { Math.PI, Math.E, Math.sqrt(a: 2), Math.sqrt(a: 3), Math.sqrt(a: 5), Math.log(a: 2),
11     Math.log10(a: 3) };
12
13 private static void sumValues(double[] arr, int arrayNumber) {
14     // Only make edits between the designated "Start" and "End" comments
15     printScenario2ArrayInfo(arr, arrayNumber);
}

Problems Output Debug Console Terminal Ports
PS C:\Users\ceray\ja692-itt14-001-f2026> java M2.Scenario2
Running Scenario 2 for [ja692] [2026-10-02T19:29:01.942039100]
Objective: Print out the total sum of the passed array
Array 1: Original Array: [0.1, 0.2, 0.3, 0.4, 0.5, 0.6]
Total Raw Value: 2.1
Total Modified Value: 2.10

Array 2: Original Array: [1.0000001, 1.0000002, 1.0000003, 1.0000004, 1.0000005]
Total Raw Value: 5.0000015
Total Modified Value: 5.00

Array 3: Original Array: [0.3333333333333333, 0.6666666666666666, 1.3333333333333333, 2.6666666666666665, 2.6666666666666665]
Total Raw Value: 7.666666666666666
Total Modified Value: 7.67

Array 4: Original Array: [1.0E16, 1.0, -1.0E16, 2.0, -2.0, 1.0E-16]
Total Raw Value: 1.0E-16
Total Modified Value: 0.00

Array 5: Original Array: [3.141592653589793, 2.718281828459045, 1.4142135623730951, 1.7320508075688772, 2.23606797749979, 0.6931471805599453, 0.47712125471966244]
Total Raw Value: 12.412475264770208
Total Modified Value: 12.41

Completed Scenario 2 for [ja692] [2026-10-02T19:29:01.973836400]
PS C:\Users\ceray\ja692-itt14-001-f2026>
```

Scenario 2 explanation

Response prompt

Explain the `M2-Homework` solution's total and two-decimal representation. Independently check the math for the starter inputs, compare expected and observed results, and investigate the discrepancy and its likely cause.

Requirements and potential deductions

- The response explains the raw-sum calculation used in the linked code. (up to -0.25 points)
- The response explains how the display is represented with two decimal places. (up to -0.1 points)
- The response compares the observed arithmetic with an independently checked expected result and investigates the discrepancy and its likely cause. (up to -0.2 points)

Submitted response

The code finds the initial total by going through every value in the array, adding each value to the total in the code. From there, specifically `String.format("%.2f", total)` to print out the total values in string values as an affirmative answer.

Scenario 3 source - positive conversion

Edit `bePositive` to make each value positive, convert it back to its original data type, and place it in the corresponding output array slot.

URL prompt

Submit the GitHub URL that opens `M2/Scenario3.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `bePositive` makes each numeric value positive. (up to -0.25 points)
- Each converted result has its original data type. (up to -0.25 points)
- Each result is stored in the corresponding output array slot. (up to -0.2 points)

<https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/M2-Homework/M2/Scenario3.java>

Scenario 3 output

Image evidence prompt

Upload one readable image of the complete `Scenario3` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output shows the expected positive results. (up to -0.35 points)
- Each displayed result corresponds to the original array position. (up to -0.2 points)

The screenshot shows a Visual Studio Code editor with the file `Scenario3.java` open. The code defines a class `Scenario3` that extends `BaseClass` and implements a static method `bePositive(Object[] arr, int arrayNumber)`. The method iterates through the array, converting each element to its absolute value while preserving its original data type (Integer, Double, Float, or String). The output is displayed in the terminal window at the bottom, showing the original array and the converted array for five different test cases.

```
public class Scenario3 extends BaseClass {
    private static void bePositive(Object[] arr, int arrayNumber) {
        //18/2/26
        //To solve this problem, it's a little tricky. But I would basically need to loop through each value and see their datatype
        //make the values positive, and then keep in the output using that same type.
        Object[] output = new Object[arr.length];
        // Start Solution Edits
        for (int i = 0; i < arr.length; i++){
            if (arr[i] instanceof Integer){
                int value = (Integer) arr[i];
                output[i] = Math.abs(value);
            } else if (arr[i] instanceof Double) {
                double value = (Double) arr[i];
                output[i] = Math.abs(value);
            } else if (arr[i] instanceof Float) {
                float value = (Float) arr[i];
                output[i] = Math.abs(value);
            } else if (arr[i] instanceof String) {
                String text = (String) arr[i];
            }
        }
    }
}
```

Output:

```
3.14159265358979[D], 2.718281828459[D], 1.61803398875[D], 0.5772156649[D], 1.0E-7[D], 1000000.0[D]
Array 3: Original Array:
1.1[F], -2.2[F], 3.3[F], -4.4[F], 5.5[F], -6.6[F], 7.7[F], -8.8[F]
Output:
1.1[F], 2.2[F], 3.3[F], 4.4[F], 5.5[F], 6.6[F], 7.7[F], 8.8[F]
Array 4: Original Array:
123[S], -456[S], 789.01[S], -234.56[S], 0.00001[S], -999999999[S]
Output:
123[S], 456[S], 789.01[S], 234.56[S], 0.00001[S], 999999999[S]
Array 5: Original Array:
-1[I], 1[I], 2.0[F], -2.0[D], 3[S], -3.0[S]
Output:
1[I], 1[I], 2.0[F], 2.0[D], 3[S], 3.0[S]
Completed Scenario 3 for [ja692] [2026-10-02T20:23:23.326675600]
PS C:\Users\ceray\ja692-it114-001-f2026>
```

Scenario 3 explanation

Response prompt

Explain the conversion and index decisions in your `M2-Homework` solution.

Requirements and potential deductions

- The response explains how the linked code preserves the original type. (up to -0.2 points)
- The response explains how the linked code selects the corresponding output position. (up to -0.35 points)

Submitted response

It basically loops through each value in the array and then checks what data type it is. I used all data types, such as Integer, Double, Float, and String.

and Float, converting the original type and using the absolute (abs) function to make it positive. For String was different, check whether the text starts with a negative sign and remove it if needed. After that process, all of the outputs will be saved in the ou array.

Scenario 4 source - text transformation

Edit `transformText` so the combined result removes non-alphanumeric characters except spaces, uses Title Case, removes leading and trailing whitespace, collapses duplicate interior spaces, and assigns the result to `placeholderForModifiedPhrase`.

URL prompt

Submit the GitHub URL that opens `M2/Scenario4.java` on your `M2-Homework` branch.

Requirements and potential deductions

- A nearby comment in the linked source supplies the required UCID, date, and solution summary. (up to -0.45 points)
- `transformText` removes non-alphanumeric characters except spaces. (up to -0.25 points)
- The source converts the phrase to Title Case. (up to -0.15 points)
- The source removes leading and trailing whitespace from the phrase. (up to -0.1 points)
- The source collapses duplicate interior spaces. (up to -0.1 points)
- The source assigns the final phrase to `placeholderForModifiedPhrase`. (up to -0.1 points)

<https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/M2-Homework/M2/Scenario4.java>

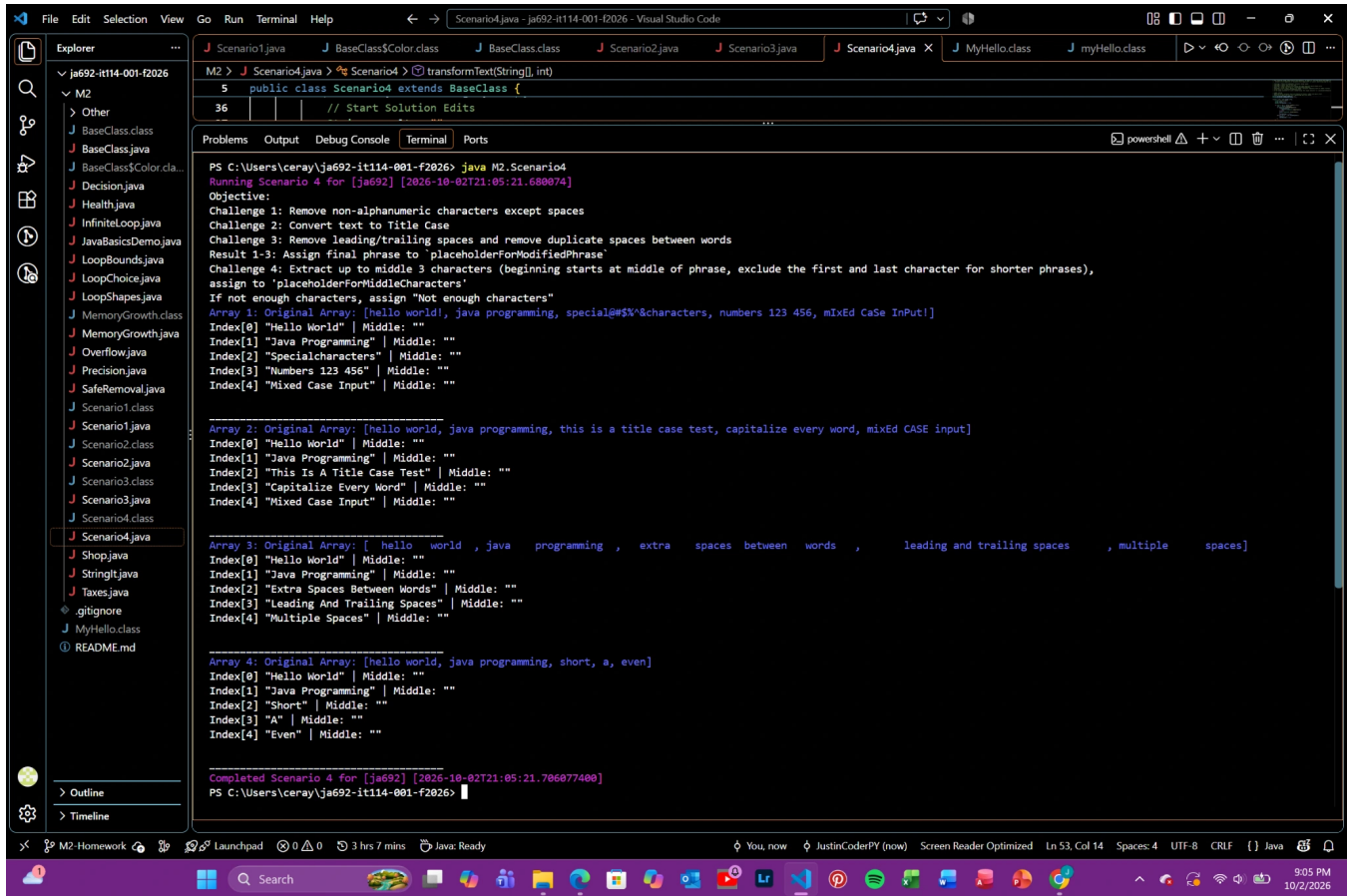
Scenario 4 output

Image evidence prompt

Upload one readable image of the complete `Scenario4` program output produced from your `M2-Homework` branch.

Requirements and potential deductions

- The output phrases contain no disallowed non-alphanumeric characters. (up to -0.25 points)
- The output phrases use Title Case. (up to -0.1 points)
- The output phrases have no leading or trailing spaces. (up to -0.1 points)
- The output phrases have no duplicate interior spaces. (up to -0.1 points)



Scenario 4 explanation

Response prompt

Explain the order of transformations and how the `M2-Homework` solution finds word boundaries.

Requirements and potential deductions

- The response explains the sequence of text-processing operations in the linked code. (up to -0.35 points)
- The response explains how the linked code finds word boundaries. (up to -0.2 points)

Submitted response

The code first removes any special characters while also keeping spaces. Then it removes extra spaces by trimming the begin and end and replacing repeated spaces with one space. Then they are split into words, and only the first letter.

Reflection

This section is worth one point total: submit your pull-request URL and separate WakaTime project/file evidence, then answer the three reflection questions.

Pull request and staged history

URL prompt

Submit the GitHub URL for the pull request from `M2-Homework` into `main`; its Commits tab must show the assignment's staged history.

Requirements and potential deductions

- The pull request targets `main` from `M2-Homework` in the personal repository. (up to -0.05 points)
- Its Commits tab shows the starter baseline, planning-comment commits made before solutions, and subsequent solution commits. (up to -0.15 points)

<https://github.com/JustinCoderPY/ja692-it114-001-f2026/pull/5>

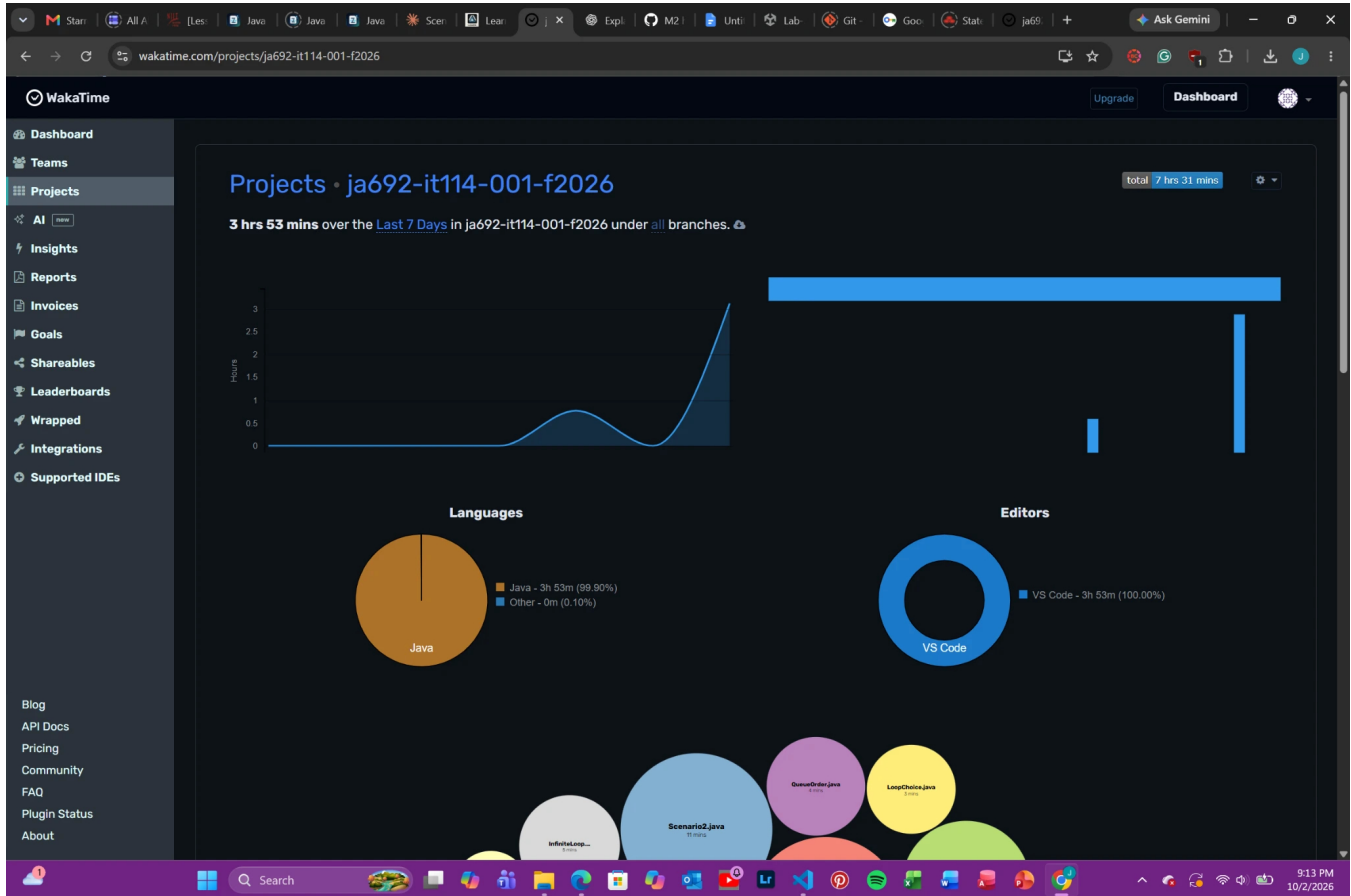
WakaTime project activity

Image evidence prompt

Upload one readable WakaTime Projects image showing overall activity for your personal IT114 repository during the period when you worked on `M2-Homework`.

Requirements and potential deductions

- The image identifies the personal repository and shows overall assignment activity without exposing an API key; duration and graphs are not graded. (up to -0.1 points)



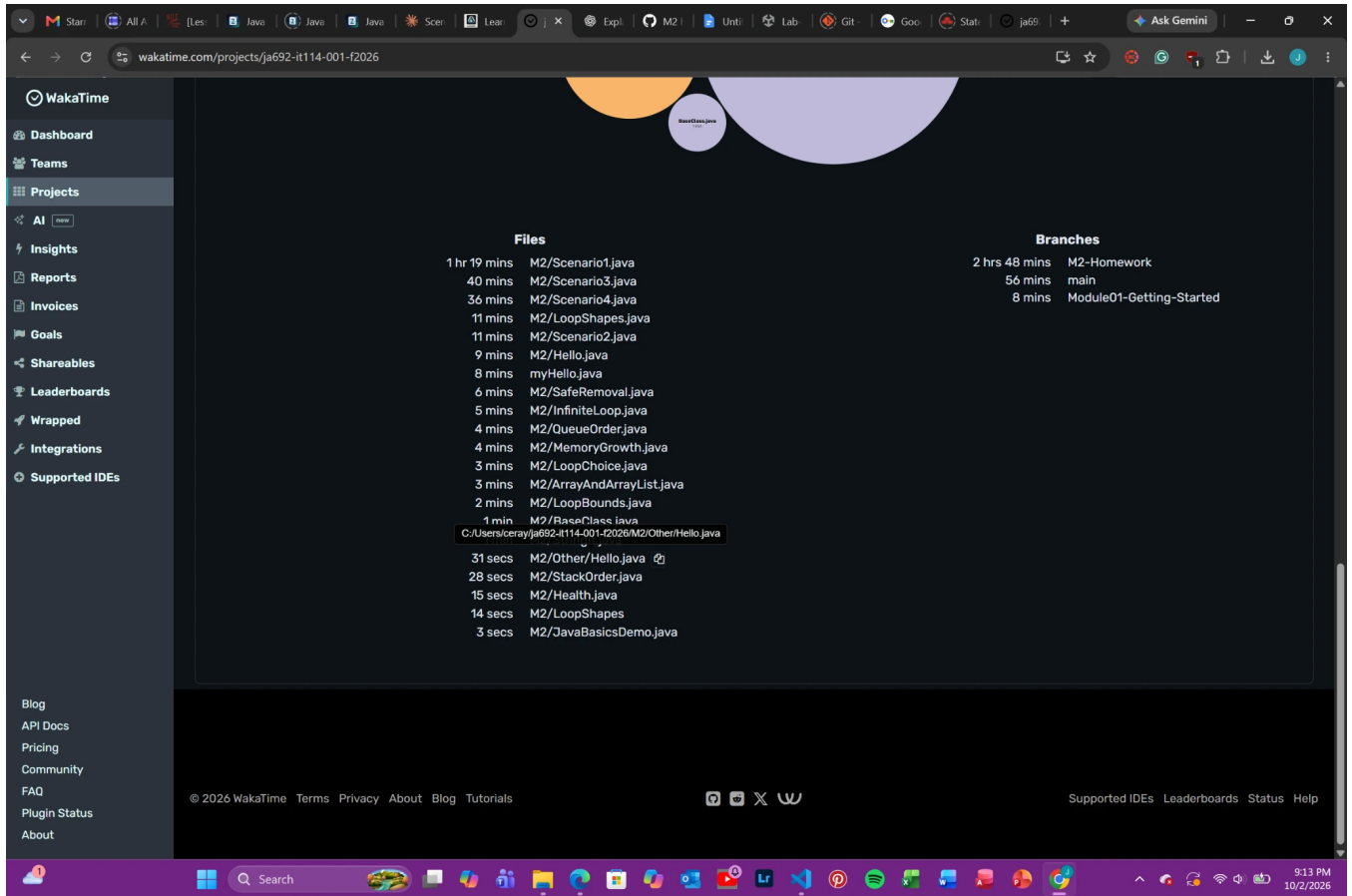
WakaTime file activity

Image evidence prompt

Upload one readable WakaTime image showing Java file activity during the period when you worked on `M2-Homework`.

Requirements and potential deductions

- The image shows relevant Java file activity without exposing an API key; duration and graphs are not graded. (up to -0.1 points)



Learned or verified

Response prompt

What did you learn or verify while completing the work on `M2-Homework`?

Requirements and potential deductions

- The response identifies something learned or verified through specific assignment experience. (up to -0.2 points)

Submitted response

I learned that sometimes I needed to do research beyond the material covered by the professor in the presentation. Some of the scenarios helped me learn new ideas that weren't explained in class, so I had to look into those same concepts and figure out how to apply them myself.

Easiest part

Response prompt

What was easiest or most straightforward in the work on `M2-Homework`, and why?

Requirements and potential deductions

- The response identifies the easiest or most straightforward part with a concrete reason. (up to -0.2 points)

Submitted response

The easiest part was definitely scenario 2. Once I understood that I just needed a simple loop through the array, adding each value to a total, and then format them to a 2 decimal point, it was pretty simple.

Most difficult or uncertain part

Response prompt

What was most difficult or uncertain in the work on `M2-Homework`, or still needs work?

Requirements and potential deductions

- The response identifies a concrete difficulty, uncertainty, or remaining next step. (up to -0.2 points)

Submitted response

The hardest part of this homework was figuring out where to find the information needed to be known when it wasn't covered in lessons. I had to learn and understand new Java features and think outside the box about how they apply to this assignment. C
good example is scenario 3, since I needed to know the different data types while also printing the same value as before.

Optional Extra Credit

Scenario 4 challenge 4 is optional and can add up to one extra-credit point. Submit its source, output, and explanation below if you attempt it; the ordinary maximum remains ten points.

Scenario 4 middle-character source

Extra-credit task

Complete optional challenge 4 in `transformText`: assign up to three middle characters to `placeholderForMiddleCharacters`, excluding the first and last characters for shorter phrases, or assign `Not enough characters` when appropriate.

URL prompt

Only if you attempted extra credit, submit the same `M2/Scenario4.java` GitHub URL you provided for Scenario 4 source, on `M2-Homework`, now including the middle-character solution.

Extra credit

Requirements and potential deductions

- The linked source assigns the requested middle-character extraction to `placeholderForMiddleCharacters` when characters remain. (up to -0.3 points)
- The linked source assigns `Not enough characters` when the phrase is too short for the extraction. (up to -0.3 points)

No URL submitted.

Scenario 4 middle-character output

Extra-credit task

Image evidence prompt

Optionally upload one readable image of the optional `Scenario4` output produced from `M2-Homework`.

Extra credit

Requirements and potential deductions

- The output shows the requested middle-character result for the applicable starter inputs. (up to -0.1 points)
- The output shows `Not enough characters` for the too-short starter input. (up to -0.1 points)

Warning at submission: minimum-items

No image submitted.

Scenario 4 middle-character explanation

Extra-credit task

Response prompt

Optionally explain how the `M2-Homework` code chooses the middle characters and handles a value that is too short.

Extra credit

Requirements and potential deductions

- The response explains the extraction decision used in the linked code. (up to -0.1 points)
- The response explains when the linked code uses the too-short fallback. (up to -0.1 points)

Submitted response

No response submitted.